Jahangirnagar University
জাহাঙ্গীরনগর বিশ্ববিদ্যালয়

Department of Computer Science and Engineering

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2nd Year 1st Semester (Hons)

Course Title: Digital Logic Design Laboratory

Course Code: CSE-254

Experiment No: 11

Experiment Title: Design and implement 2 bit Asynchronous binary counter

Submitted To:

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Experiment No: 11

Experiment Name: Design and implementation of a 2 bit synchronous counter

Objectives:

To design and implement a 2-bit synchronous counter using JK flip flops where all flip flops are clocked simultaneously and verify the glitch free binary counting sequence.

Required components:

Digital circuit board, IC 7476, IC 7408, wires.

Theory:

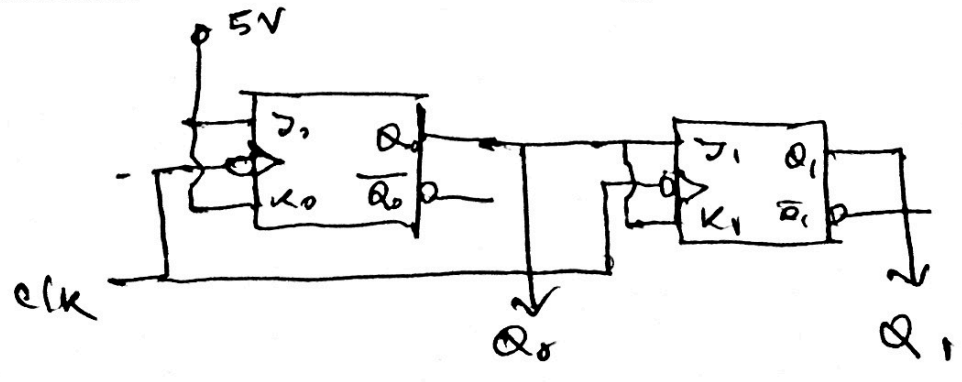
A synchronous counter is a sequential circuit where all flip flops are triggered simultaneously by the same clock signal.

For a 2bit synchronous counter,

LSB toggles, when $J_0 = K_0 = 1$

MSB " , " $J_1 = K_1 = Q_0$

Circuit Diagram:



Truth Table:

clk	Q_1	Q_0
0	0	0 ↓
1	0 ↓	1 ↓
2	1 ↓	0 ↓
3	1 ↓	1 ↓
4	0 ↓	0

$J_0 = K_0 = 1$

$J_1 = K_1 = Q_0$

Procedure:

- Place all the ICs on the bread board.
- Connect the J_1K_1, J_0K_0 correctly
- Match the output with the sequence.

Result:

- The circuit successfully counted $00 \rightarrow 01 \rightarrow 10 \rightarrow 11 \rightarrow 00$
- Both flip flop changed state simultaneously
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Discussion:

The synchronous counter worked as expected demonstrating the binary counting.